

Discussion Unit:

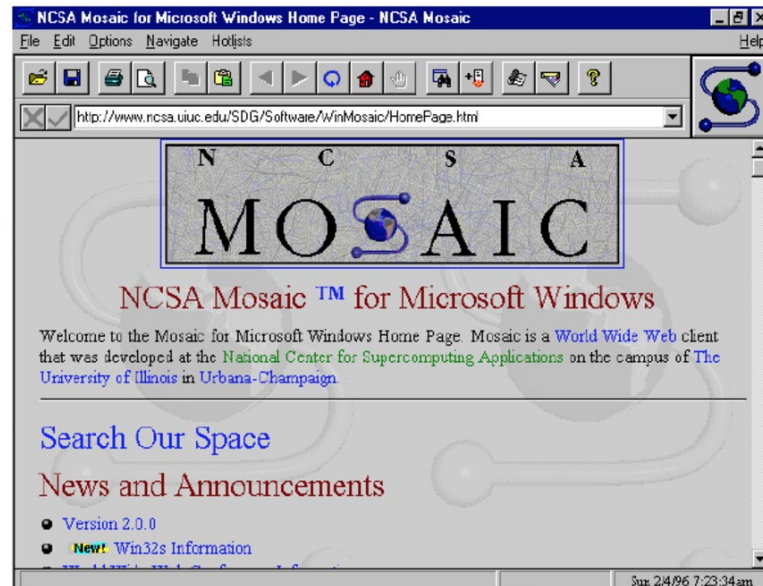
DSE Studio, Overview,
(planned use)

Introduce DSE Studio

- Understand where Studio fits in the DSE portfolio
- Understand when/how you may choose to use DSE Studio

Discussion Lab:

Matching pairs – Match the attributes on the right with the areas on the left



NCSA Mosaic, or simply Mosaic, is the web browser that popularized the World Wide Web and the Internet. It was also a client for earlier internet protocols such as File Transfer Protocol, Network News Transfer Protocol, and Gopher. The browser was named for its support of multiple internet protocols.

Source: [https://en.wikipedia.org/wiki/Mosaic_\(web_browser\)](https://en.wikipedia.org/wiki/Mosaic_(web_browser))

Discussion Lab: Clients to DSE version 6 ?

DATASTAX

DSE

Not DSE



Ops Center

Studio

Dev Center

DSBulk

DSFiber (H-NAS only)

GraphLoader

DSE supported drivers

Open source drivers

Box of Tissues

Source: http://www.spencer1984.com/my_models/beverly-hillbillies.php

End of Discussion Lab:



DataStax Studio

An intuitive and collaborative tool for developers, designed to help your Cassandra Query Language (CQL), DSE Graph, and Spark SQL language development. Easily collaborate between developers and analysts with self-documenting notebooks.

Visually Interact with Database Schemas

Visual representations of your schema objects help you understand your models and build them quicker. Navigating and managing complex database schemas can be difficult. To help you easily identify design or model bottlenecks, optimize queries and save time, DataStax Studio also allows you to interact with multiple clusters.

Tune Queries for Faster Performance

DataStax Studio features query profiling for CQL, Spark SQL and Gremlin, providing diagnostic information and execution plan details that let you understand bottlenecks and tune queries for fast performance.

Query and Explore Data with Ease

DataStax Studio features an intelligent editor that simplifies creating complex CQL, Spark SQL, and Gremlin queries. This powerful code editor provides context-aware suggestions and validation, allowing you to write queries without constantly referring to documentation or schemas.

Graph and Data Visualization

The rich set of data visualizations and numerous output formats enable you to surface insights from data and present them in interactive, publication-quality graphics. DataStax Studio features grid, JSON, and plain-text viewers. You can also use results in pie, bar, line, and area charts. Last but not least, DataStax Studio offers a specialized visual

Self-Documenting Notebooks

DataStax Studio provides an intuitive interface for developers and analysts to collaborate by mixing code, documentation, query results, and visualizations into self-documenting notebooks that can be shared with colleagues.

DataStax Loader

Quickly and easily load and unload data into and out of DSE factors faster than manual techniques.

DSE Studio:

- Based on Zeppelin notebook
- Runs locally; (lightweight) Web app
- Supports:
 - CQL (and Search)
 - Spark/SQL (w/ Always on SQL)
 - Gremlin
 - Markdown
- Import/export, w/w-o data
- Content assist (Shift-ENTER)

DSE Studio: Download, boot

Downloads: Full List

Download from
Academy

```
./server.sh
```

Studio

RESET

Title	Version Number	Download/Driver Category	Platforms	Documentation and Additional Links	Download Link
DataStax Studio 6.0 (for DSE 6)	6.0	Studio	• CentOS, RHEL, Oracle Enterprise Linux, Ubuntu, Debian, SUSE, Mac OSX*, Windows	• Documentation	• MacOSX/Linux • Windows
DataStax Studio 2.0 (for DSE 5.x)	2.0	Studio	• CentOS, RHEL, Oracle Enterprise Linux, Ubuntu, Debian, SUSE, Mac OSX*, Windows	• Documentation	• MacOSX/Linux • Windows

DSE Studio: Boot process

```
# ./server.sh
```

Starting Studio. This may take a few minutes. You will be notified here when Studio is ready.

SLF4J: Class path contains multiple SLF4J bindings.

SLF4J: Found binding in [jar:file:/opt/studio/lib/log4j-slf4j-impl-2.9.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]

SLF4J: Found binding in [jar:file:/opt/studio/tomcat.9091/webapps/api/WEB-INF/lib/log4j-slf4j-impl-2.9.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]

SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.

SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]

Studio is now running at: <http://localhost:9091>

NOTE: Studio by default will only be accessible on this machine since it is bound to localhost.

Studio is intended to be used as a desktop application. To allow remote connections(which can be a security risk) modify the httpBindAddress setting in configuration.yaml to either a publicly accessible address or 0.0.0.0

Please visit https://docs.datastax.com/en/dse/6.0/dse-dev/datastax_enterprise/studio/configStudio.html for more information on configuration

DSE Studio: First Steps

DataStax Studio – Mozilla Firefox

File Edit View History Bookmarks Tools Help

New Tab x DataStax Studio x +

localhost:9091

Google z Content 1 http://localhost:8081/ DataStax Studio Solr Admin Spark Master at spark:/... Understanding DSE Se... DSE 5.1 Resource Man...

≡ DataStax Studio Notebooks

Recent All Notebooks Starred Search Notebooks

+

Or drop files to import

What's New in Studio v6.0.0
default localhost
★ a few seconds ago

Working with CQL v6.0.0
default localhost
★ a few seconds ago

Working with Graph v6.0.0
default localhost
★ a few seconds ago

DSE Graph QuickStart v6.0.0
default localhost
★ a few seconds ago

Working with Spark SQL v6.0.0
default localhost
★ a few seconds ago

DSE Studio: Create new (CQL) Notebook

Working with Graph v6.0.0
default localhost

DSE Graph QuickStart v6.0.0
default localhost

Working with Spark SQL v6.0.0
default localhost

CREATE CONNECTION

Name *
my_local

Username

Host/IP (comma delimited) *
127.0.0.1

Password

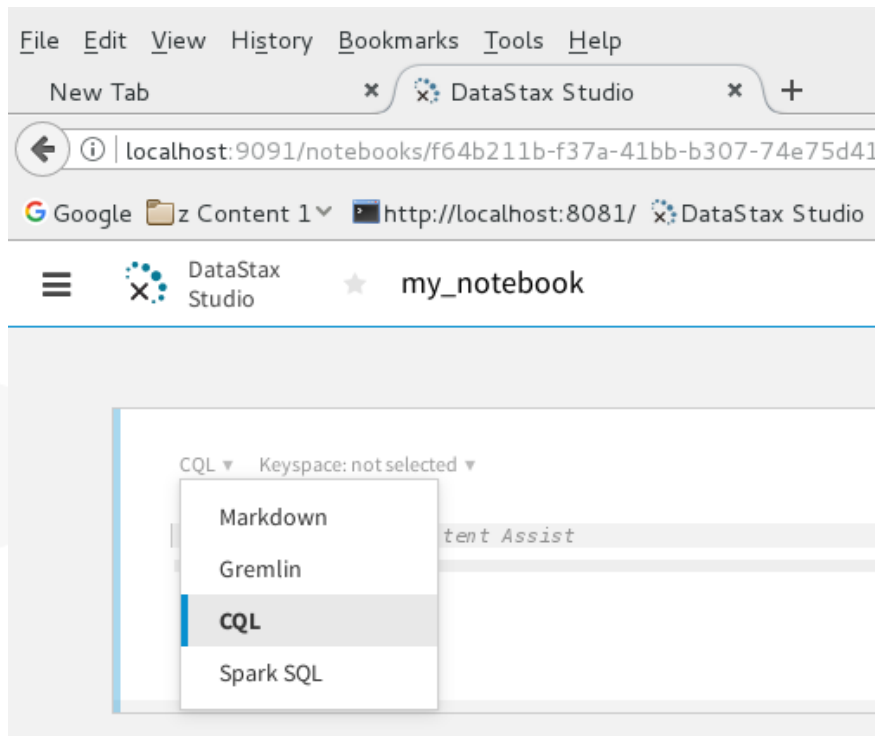
Port *
9042

☐ Use SSL

Save **Cancel**

Connected successfully **Test**

DSE Studio: Language within a Cell



DSE Studio: Running CQL

CQL ▾ Keyspace: not selected ▾

```
DROP KEYSPACE ks_7443;  
CREATE KEYSPACE ks_7443 WITH REPLICATION =  
  {'class': 'SimpleStrategy', 'replication_factor': 1};  
USE ks_7443;
```

Success - No Data Returned

Success. 0 elements returned. Duration: 1.882 s.

▶ **Execute with CL.ONE**

▶ Execute with CL.QUORUM

▶ Execute with CL.ALL

▶ Execute with CL.ONE.TRACE

⚙ Manage Configurations

DSE Studio: Standard CQL

CQL Keyspace: ks_7443

```
CREATE TABLE t1
(
  col1          TEXT,
  col2          TEXT,
  col3          TEXT,
  col4          TEXT,
  col5          TEXT,
  col6          TEXT,
  col7          TEXT,
  col8          TEXT,
  PRIMARY KEY ((col1, col2)) );

INSERT INTO t1
(col1, col2, col3, col4,
 col5, col6, col7, col8)
VALUES ('aaa', 'aaa', 'Davie',
 'aaa', 'aaa', 'aaa', 'aaa', 'aaa');

INSERT INTO t1
(col1, col2, col3, col4,
 col5, col6, col7, col8)
VALUES ('bbb', 'bbb', 'Elizabeth',
 'bbb', 'bbb', 'bbb', 'bbb', 'bbb');
```

Success - No Data Returned

Success. 0 elements returned. Duration: 1.135 s.

DSE Studio: Chart Types

CQL ▾ Keyspace: ks_7443 ▾

| DESCRIBE TABLE t1;



NOTE: DESCRIBE is a feature of cqlsh and is not a valid CQL statement.

```
CREATE TABLE ks_7443.t1 (  
    col1 text,  
    col2 text,  
    col3 text,  
    col4 text,  
    col5 text,  
    col6 text,  
    col7 text,  
    col8 text,  
    PRIMARY KEY ((col1, col2))  
) WITH read_repair_chance = 0.0  
    AND dclocal_read_repair_chance = 0.1  
    AND gc_grace_seconds = 864000  
    AND bloom_filter_fp_chance = 0.01  
    AND caching = { 'keys' : 'ALL', 'rows_per_partition' : 'NONE' }  
    AND comment = ''  
    AND compaction = { 'class' : 'org.apache.cassandra.db.compaction.S
```

DSE Studio: More Chart Types

CQL Keyspace: ks_7443

```
select * from t1;
```



index ↑	col1	col2	col3
0	bbb	bbb	Elizabeth
1	aaa	aaa	Davie

DSE Studio: Subject to the underlying DSE

CQL ▾ Keyspace: ks_7443 ▾

```
■ CREATE SEARCH INDEX ON ks_7443.t1 WITH COLUMNS col1,  
  * { excluded : true };
```

Execution Error

An error occurred on line 1:
Search statements are not supported on this node
(Hint: Ctrl-L to toggle line numbers.)

DSE Studio: Subject to the underlying DSE

Spark SQL

Database: ks_7443



Cell cannot be executed because DSE Spark Analytics workload is not enabled or AlwaysON SQL service is not enabled and running in connection [my_local](#). [Learn more](#)

Control+Space for Content Assist

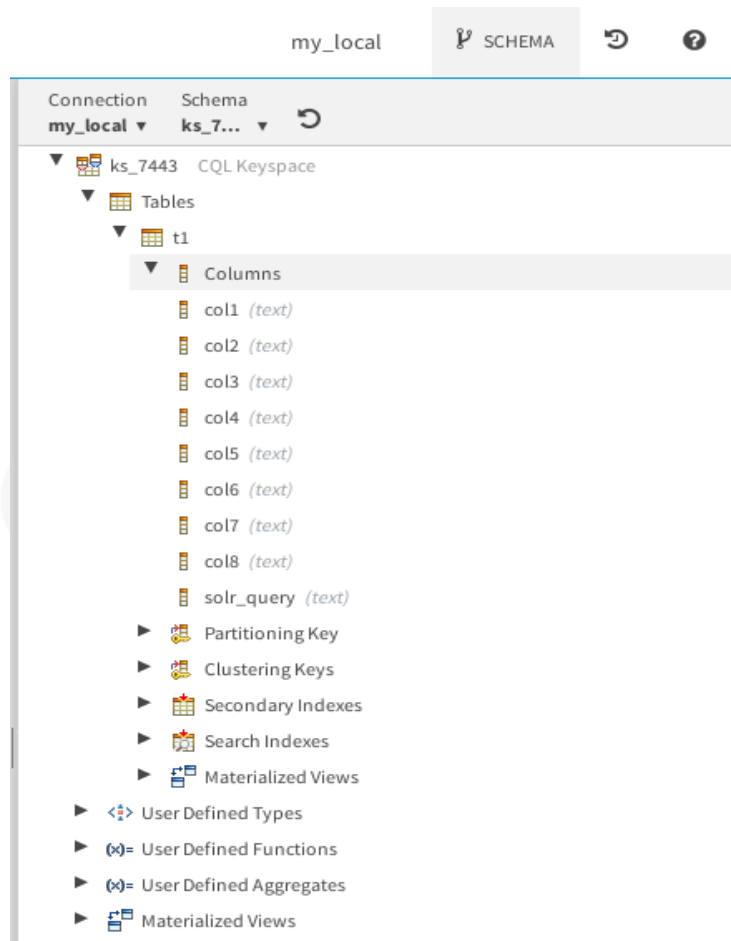
DSE Studio: Successful Spark/SQL

Spark SQL ▾ Database: ks_18 ▾

```
SELECT COUNT(*) v_groupCount, col3
FROM t1
WHERE ((col3 >= 'A') AND (col3 <= 'Z'))
GROUP BY col3
ORDER BY v_groupCount
LIMIT 10;
```



index ↑	v_groupCount
0	4



DSE Studio: Schema Explorer

